FAQIH ALAM

TANGERANG SELATAN, INDONESIA

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PERSONAL PROFILE

Bachelor's degree in Electrical Engineering from Telkom University (GPA 3.45), with strong foundations in **embedded systems**, **IoT applications**, and **automation engineering**. Skilled in **microcontroller programming (Arduino, ESP32, Raspberry Pi), IoT protocols (MQTT, HTTP), PLC programming, and hardware-software integration**. Experienced in developing **embedded-based projects such as smart door locks, automation systems, robotics, and face recognition systems**. Hands-on knowledge in **Python, C/C++, and hardware interfacing with sensors and actuators**. Practical exposure as a Service Engineer in UPS systems, ensuring power reliability at data centers. Highly motivated to contribute as an Embedded Engineer, combining programming, circuit design, and system integration to build reliable, scalable embedded solutions.

WORK EXPERIENCE

Service Engineer at PT Deltasondo Raya Sejahtera | July 2024 – August 2025

- Conducted **preventive and corrective maintenance** on Uninterruptible Power Supply (UPS) systems.
- Perform Quarterly Preventive Maintenance of Uninterruptible Power Supply (UPS) at Data Center Indonesia, PT. KCI, PT. Datacomm, PT. PDG.
- Performed **Startup UPS, battery installation, load testing, and monitoring** to ensure system reliability.
- Perform 600KVA UPS Rotation (Startup/Shutdown UPS) at Data Center Indonesia MM2100

ELITIA Research Laboratory | 2021-2022

- Researching about industrial automation field using ladder diagram programming language with CX Programmer; using Omron Programmable Logic Controller (PLC); making Human Machine Interface (HMI) with CX Designer.
- Researching about internet of things (IOT) field; using Arduino, Raspberry Pi and NodeMCU ESP 8266 or 32 as microcontroller; using MQTT & HTTP as protocol; developing mobile applications with Kodular & MIT APP Inventor.
- Teaching college students about industrial automation including PLC and Internet of Things application in industrial field.
- Create Workshop for college students for making Internet of Things concept work.
- Teaching college students about Machine Learning including Perform Exploratory Data Analysis, Feature Engineering, Building Model, Evalute Model and Deploy Model.

Internship at Telkom Jakarta Barat | June 2020 – August 2020

- Maintaining Metropolitan Area Network for internet use; fixing or create new route cables.

PROJECT

Service Engineer at PT Deltasondo Raya Sejahtera

- Installed and replaced VRLA batteries for 600 kVA UPS at Data Center Indonesia, MM2100.
- Installed and replaced VRLA batteries for 10 kVA UPS at Bank IBK.
- Installed and replaced VRLA batteries for 80 kVA UPS at Mitra Keluarga Hospital.
- Installed an 11 kVA Eaton UPS at PT Indogrosir, Pulo Gadung.

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Student of Telkom University Electrical Engineer

- Undergraduate Thesis – Home Security System: Built a proof-of-concept home security system with Raspberry Pi, implementing face recognition (KNN ML model) for doorlock authentication. Integrated with Telegram for IoT-based monitoring and control of the doorlock and camera.
- Developed a Machine Learning Model for Skin Moisture Classification.
- Developed a Machine Learning Model for ID Card (KTP) Processing and Classification.
- Designed and implemented a conveyor-based automation system to sort boxes by height, integrated proximity sensors and actuators, with control logic programmed on a PLC.
- Built an IoT-based system using Telegram for Monitoring & Controlling doorlock system.
- Designed and built Various IoT system with Nodemcu/Arduino with multiple sensors.
- Built a Line Follower Robot and automated control systems for educational use.
- Created a Door Lock System using Raspberry Pi, IoT, Telegram, and Face Detection.

EDUCATION

- Bachelor's Degree in Electrical Engineering from Telkom University (2018-2022)
- BANGKIT ACADEMY GRADUATE (2021)
  Machine Learning Path Student
  - Learned about clustering, predicting, and classifying data through platforms such as Coursera and Qwiklabs.
  - Built an application to extract text and identify information from ID cards, such as names and places of birth, utilizing machine learning algorithms, mobile app development, and cloud computing.
- Graduate in Science Major from SMAN 90 Jakarta Barat (2015-2018)

CERTIFICATION & LICENCSE

- Indobot – Basic Internet of Things
- Linkedin – Programming Foundations: Fundamentals
- Coursera – Crash Course on Python
- Coursera – Google IT Support
- Coursera – Google IT Automation with Python
- Coursera – Mathematics with Machine Learning

ADDITIONAL SKILLS AND INTERESTS

- Embedded Systems Development: PCB/Circuit Design, Arduino, ESP, Raspberry Pi, Sensors, Actuators, C/C++/Python, Real-time programming, IoT protocols
- Automation & Control Systems: PLC Programming (Ladder Diagram, Omron CX-Programmer), HMI (CX-Designer).
- Data Analysis & Business Intelligence: Python (Pandas, NumPy, Matplotlib, Scikit-learn), Excel, SQL (basic).
- Machine Learning & AI: OpenCV, Image classification, Supervised & Unsupervised model training, overfitting handling.
- Power Systems: Uninterruptible Power Supply (UPS) installation and maintenance.
- IoT Development: Arduino, Raspberry Pi, NodeMCU, MQTT, HTTP protocols.

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- Problem-Solving & Communication: Technical reporting, teaching, stakeholder presentations.
- Languages: Fluent in Bahasa Indonesia and English.

ORGANIZATION EXPERIENCE

- Chief of UKM Tae Kwon Do Telkom University in 2020.
- Part of UKM Tae Kwon Do Telkom University as Logistic Division member 2019
- Part of Keluarga Mahasiswa Teknik Elektro at Telkom University.
- Part of ELITIA Laboratory as Logistic Chief 2021-2022.
- Part of ELITIA Laboratory as IoT Assistant Practicum Division 2021-2022.